

Reproductive ecology of Caribbean reef corals

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Abstract. The last decade has seen a resurgence of interest in the processes of sexual reproduction by scleractinian reef corals. Earlier investigations had focused fortuitously on brooding (planulating) species, which resulted in the general misconception that brooding was the main form of larval development of reef corals. More recent work on Indo-Pacific species has shown broadcast spawning and short annual reproductive periods to predominate. This report presents the reproductive patterns of eleven Caribbean coral species and attempts to explain the adaptive features and selective pressures that have led to the evolution of the four reproductive patterns described to date: (a) hermaphroditic broadcasters; (b) gonochoric broadcasters; (c) hermaphroditic brooders; (d) gonochoric brooders. Both (a) and (b) correlate with large colony size and short annual spawning periods; and (c) and (d) correlate with small colony size, multiple planulating cycles per year, and occupation of unstable habitats. Selection for outcrossing between long-lived individuals is proposed as the reason for gonochorism and for synchronous spawning of hermaphroditic broadcasters, and also for the large amount of sperm produced by hermaphroditic brooders. Selection for high rates of local recruitment is proposed as the force behind the evolution of brooding by species inhabiting unstable habitats and suffering high rates of adult mortality.

Introduction

Reef coral communities are diverse and complex. At present, there is much ongoing research aimed at understanding the factors that contribute to and control coral community structure. Physical factors, such as physical disturbances, temperature, salinity, desiccation, as well as biological factors, such as predation and competition all have been shown to be important – to varying degrees, and in varying combinations – on different reefs and in different reef habitats (see recent review by Done 1983). Presumably coral species have adapted and evolved in re-

sponse to these factors, and their life history characteristics should reflect the selection of traits which are adaptive to the various situations. Growth form and reproduction are two traits that can vary greatly between species, and thus may be supposed to have been acted upon by natural selection.

Differences between corals in growth form characters, such as colony size and shape, are rather obvious even to the casual observer. Branching species, which often assume large dimensions, can dominate extensive areas of reef (Gilmore and Hall 1976; Glynn 1976); foliaceous species predominate along vertical surfaces (Bak and Luckhurst 1980); massive species are structurally restricted to areas of limited slope (Goreau 1959); and, numerous smaller heading, branching and encrusting species have a wide distribution within the reef and vary in abundance from rare to quite numerous. The various colony shapes and sizes are thought to have adaptive value. Large size and massiveness increase resistance to all but the most devastating of physical or biological disturbances, and imply great longevity. Branching morphology, combined with fast growth rate, can result in longevity and the dominance of large areas (as in the case of *Acropora cervicornis*), but may also result in greater susceptibility to physical disturbance (Woodley et al. 1981; Highsmith 1982). Small size implies either an early cessation of growth (determinate growth), slow growth or high colony mortality. When due to the latter cause, high abundances must be associated with high recruitment rates, which in turn require high success in reproduction.

While the short term population dynamics of large, massive or fast-growing, branching species may not be greatly influenced by rates of recruitment of new sexually produced individuals, their ability to colonize new areas, to recolonize old catastrophically disturbed areas, and in general, their ability to persist as a species depends on some degree of success in sexual reproduction.

Sexual reproduction consists of a sequence of related events, each of which can be individually modified by

natural selection. These include gametogenesis, release of the gametes from parental tissues, fertilization and embryogenesis. Initial recruitment of larvae into a population are also part of the reproductive sequence because much of the selective pressure will take place during these stages (Vance 1973; Crisp 1977). The timing, frequency, duration, intensity and spatial distribution of these events determines whether the reproductive effort is successful. The net result of these events, settlement, is what is usually measured as the birth rate of the population. A reproductive strategy is the sum of the particular ways by which a species accomplishes the reproductive events outlined above.

In contrast with growth traits, reproductive traits are not obvious, and take a considerable amount of effort to discern. Only within the last decade or so has there been a resurgence of interest in sexual reproduction by reef corals. Most of the earlier and some of the more recent reports on sexual reproduction of scleractinian reef corals focused on those species that brooded their embryos to the planula stage before release (= planulation) (e.g. Duerden 1902, 1904; Marshall and Stephenson 1933; Harrigan 1972; Rinkevich and Loya 1979 a, b and many others; see review by Fadlallah 1983). Probably because of this, a misconception arose that brooding was the dominant form of reproduction in these animals. More recently, it has been demonstrated that most corals do not brood their larvae, but are, in fact, broadcasters (e.g. Szmant-Froelich et al. 1980; Kojis and Quinn 1981 a, b; Fadlallah 1982; Fadlallah and Pearse 1982; Krupp 1983; Harriott 1983; Harrison et al. 1984; Babcock 1984; Shlesinger and Loya 1985; Willis et al. 1985; Harrison 1985).

A wide variety of patterns of sexual reproduction have been reported recently for Indo-Pacific (see summary in Fadlallah 1983; Harrison et al. 1984) and Caribbean species (von Moorsel 1983; Szmant-Froelich 1984; unpublished results summarized below, detailed accounts on each species will be reported elsewhere). I will try to relate the observed reproductive traits and patterns to growth characteristics of the species, and I will suggest reasons why particular combinations of traits may be adaptive to particular ecological situations. The ultimate goal of these studies is to provide a framework to help understand the selective forces that may have resulted in the evolution of the various reproductive strategies.

Materials and methods

These reproductive studies summarized here were conducted from the Maguayes Laboratory of the Department of Marine Sciences, University of Puerto Rico, La Parguera, Puerto Rico. The La Parguera area has extensive coral reef development, with several lines of mid-shelf reefs and a deep submerged reef ("The Buoy") at the edge of the insular shelf 8 km south of La Parguera. All of the species reported for the West Indies region (Wells 1973) occur in this area.

Biweekly sampling (5 to 40 specimens for each species) was begun in November of 1982 from La India reef, and continued through November, 1983. Weekly sampling was performed from December, 1983, through May, 1984, for the one winter-breeding species (*Myceto-*

phyllia ferox) and for most of the others from mid-June through mid-September, 1984. The La India site was selected because it had an abundance of many of the coral species of interest: *Acropora palmata*, *Montastrea annularis*, *M. cavernosa*, *Diploria strigosa*, *Siderastrea siderea*, *Porites astreoides* and *Mycetophyllia ferox*. In addition, *Acropora cervicornis* and *Favia fragum* were studied on the nearby San Cristobal reef and *Dendrogyra cylindrus* on El Mario reef. All samples were fixed in the field with Helly's fixative and processed for histological examination by standard techniques. Sections were stained with Heidenhain's azocarmine-aniline blue. Both longitudinal and horizontal sections were made for most species. Three to four slides of each orientation were examined with a compound light microscope to examine the distribution of gonadal tissue within the polyps. Gametes present were classified as to stage of development according to criteria based on cell size and morphology (see Szmant-Froelich et al 1980). Temporal plots of gametogenic stages were made to determine the timing of the gametogenic cycles (Fig. 1).

Results

All species studied except for *Favia fragum* were found to have annual cycles of gametogenesis. The reproductive traits and patterns of each species are summarized in Table 1. The results show that there is much variability in reproductive pattern between the species studied. Four of the species were gonochoric, the rest were annual protogynous hermaphrodites. Three of the species were hermaphroditic brooders and one (*Siderastrea radians*) was a gonochoric brooder. All of the brooders were smaller sized species and all of the large-sized species were broadcasters. The outermost polyps of most species and the terminal two cm of *A. cervicornis* branches seldom had gametes.

Examination of the range of gametocyte stages present in each colony at each sampling date show that the populations were generally synchronous in their development (Fig. 1). The greatest differences between colonies were during the earliest stages of oogenesis, with some colonies beginning gametogenesis a few weeks before others. Throughout most of the oogenic period there were many stages present simultaneously within individual gonads. There was much more synchrony during the final phases of oocyte maturation, and during spermatogenesis, so that individuals within the population were ripe at the same time (Fig. 1). Spawning patterns (as evidenced by the disappearance of mature gametocytes from the tissues; Fig. 1) differed for each species. All of the broadcasters were mature during the late summer (July–September) (Table 1), but spawning does not appear to be synchronized between species. Spawning was observed in the field for a single colony of *Diploria strigosa* 7 nights after the July/August, 1985, full moon, and preparations for spawning (consolidation of gamete bundles in upper coelenteron) for *Acropora cervicornis* on night 6 after the August, 1984, full moon and on nights 7 and 8 after the July/August, 1985, full moon. Laboratory maintained colonies of *A. cervicornis* spawned on these same nights. Several colonies of *Montastrea annularis* spawned in the laboratory eight days after the September, 1984, full moon. Spawning behavior by *D. strigosa* was similar to that reported for Great Barrier Reef corals (Harrison et

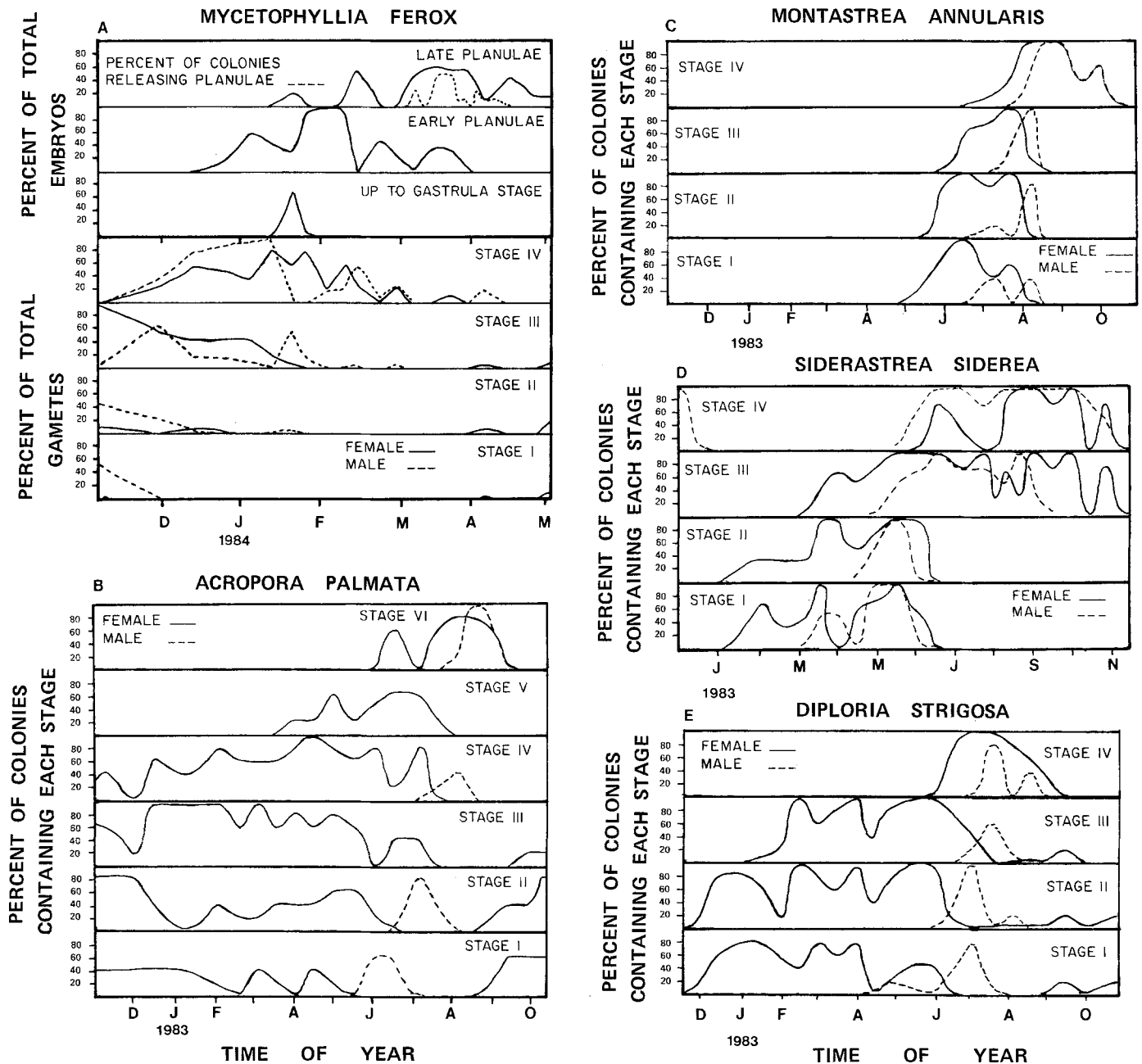


Fig. 1 A-E. Representative reproductive cycles of five of the species studied. **A** *Mycetophyllia ferox*: Period shown is from November 1983 to May 1984 during which weekly samples were taken. Stages I-IV are categories of increasing gametocyte maturity with IV being ripe. Y-axis is the percent of all the gametes (or embryos) counted for each sampling date made up of gametocytes (or embryos) of each stage. The upper box includes the percent of colonies releasing planulae during in situ planulation study. **B** *Acropora palmata*: Annual cycle of gametogenesis with VI oocytes stages and IV spermary stages. Male Stages III and IV were plotted with female Stages IV and VI respectively. Y-axis is the percent of colonies sampled each date that had the given stage in their gonads. **C** *Montastrea annularis* (**D**) *Siderastrea siderea* and (**E**) *Diploria strigosa*: Similar to plot of *A. palmata* but only four stages defined in their oogenic cycle. In all of the above, except *S. siderea*, note the presence of mature ova and sperm (Stages IV or VI) during only a very short time. For more detail on methods, see Szmant-Froelich et al. 1980, 1985

al. 1984; Willis et al. 1985): large bundles 2-4 mm in diameter of gametes were released by individual polyps over a 20-30 min period at about 2100 h. The bundles were buoyant and floated towards the surface. Several ophiuroids were crawling over the surface of the colony gorging on the gamete masses. It was a spectacular sight which would have been difficult to overlook if come across. These were the only coral spawning events ob-

served during 2 months of diving every other night from 1900-2030 hours during the late summer of 1984 and during 5 night dives beginning 4 days after the July/August, 1985 full moon.

Of the brooders, one (*Mycetophyllia ferox*) brooded during the winter-time only (January-April) (Fig. 1a), but two others had 9-12 month brooding periods. One of the latter, *Favia fragum*, had lunar gametogenic and em-

Table 1. Summary of reproductive patterns of some Caribbean reef corals from Puerto Rico

Species	Sexual mode	Egg size μm^a	Gametogenesis	Spawning season	Brooding season (planula size)
Suborder Astrocoeniida					
Fam. Acroporidae					
<i>Acropora palmata</i> (Lamarck)	Hermaphroditic protogynous	300	♀ Sept. → May ♂ May → July	Aug.	No
<i>Acropora cervicornis</i> (Lamarck)	Hermaphroditic protogynous	300	♀ Sept. → April ♂ June → July	Aug.	No
Suborder Fungiida Fam. Poritidae					
<i>Porites astreoides</i> Lesuer	Hermaphroditic simultaneous	50	♀ Year-round ♂ Dec. → June	Lunar?	Jan. → Sept. (1 mm)
Fam. Siderastreidae					
<i>Siderastrea siderea</i> (Ellis and Solander)	Gonochoric	600	♀ Feb. → June ♂ April, May	July → Sept.	No
<i>Siderastrea radians</i> (Pallas)	Gonochoric	400	?	?	Year-round?
Suborder Faviida Fam. Faviidae					
<i>Favia fragum</i> (Esper)	Hermaphroditic simultaneous	250	♂ Lunar, year round,	Day 18 after new moon planulate: days 8–16	Lunar Days 18 → 8 (1–2mm)
<i>Diploria strigosa</i> (Dana)	Hermaphroditic protogynous	400	♀ Mid Dec. → June ♂ June, July	Mid Aug.	No
<i>Montastrea annularis</i> (Ellis and Solander)	Hermaphroditic protogynous	300	♀ Mid May → July ♂ July, Aug.	Mid Aug. to Mid Sept.	No
<i>Montastrea cavernosa</i> (Linnaeus)	Gonochoric protogynous	350	♀ Late Oct. → July ♂ Mid May, June	Late Aug.	No
Fam. Trochosmiliidae					
<i>Dendrogyra cylindrus</i> (Ehrenberg)	Gonochoric protogynous	300	♀ May–Aug. ♂ Mid June → Aug.	Mid Aug ?	No?
Fam. Mussidae					
<i>Mycetophyllia ferox</i> Wells	Hermaphroditic protogynous	300	♀ Mid April → Aug. ♂ Nov., Dec.	Dec.–Jan; Planulation: mid Feb., March Lunar?	Jan. → March (3 mm)

^a Maximum diameter measured in histological sections**Table 2.** Estimates of average annual egg volume production per cm^2 surface area for various Caribbean reef corals. E=external; B=brooding

	Egg diameter μm	Number of eggs per gonad	Number of gonads per polyp	Number of polyps per cm^2 ^a	Number of reproduction cycles/year	Total egg volume $\text{mm}^3/\text{cm}/\text{year}$	Type of development ^b
<i>Acropora cervicornis</i>	300	6–8	4–6	25	1	12.25	E
<i>Porites astreoides</i>	50	2	6–10	18	9	0.16	B
<i>Siderastrea siderea</i>	600	8–10	22–32	9	1	116.63	E
<i>Favia fragum</i>	250	1–2 ^b	9–22	12	12	27.36	B
<i>Diploria strigosa</i>	400	8–10	22–32	2	1	18.10	E
<i>Montastrea annularis</i>	300	5–14	12	12	1	24.48	E
<i>Montastrea annularis</i>	350	12	24	2	1	12.94	E
<i>Dendrogyra cylindrus</i>	300	10–30	8 ^c	3	1	6.74	E
<i>Mycetophyllia ferox</i>	300	2–4	12–24	2	1–2	1.51	B

^a Estimated from species descriptions in Smith (1971)^b Mature oocytes at a time; many immature ones present also^c Per cm^2 ; difficult to distinguish polyp boundaries

bryogenic cycles all year long (Szmant-Froelich et al. 1985).

There were no defined patterns within systematic groups. Congeneric species differed with regard to such fundamental features as sexuality and mode of embryogenesis. Therefore, differences in patterns of reproduction may be useful traits with which to distinguish between morphologically similar species (e.g. *S. siderea* and *S. radians*, which have recently been synonymized; Zlatarski 1982). More detailed accounts of the reproductive patterns of each species will be presented elsewhere.

Discussion

Three important conclusions can be drawn from these recent studies: (1) contrary to some suggestions (Bak and Engel 1979), Caribbean reef corals do have well-defined seasonal patterns of sexual reproduction, even those species that are successful at vegetative reproduction; (2) the amount of resource allocated to reproduction may be considerable, as judged by the large number and size of the reproductive products (Table 2); and (3) patterns of sexual reproduction are not significantly correlated with

polyp size or morphology, as has been suggested by several previous investigators (Fadlallah 1980; Stimson 1978; Rinkevich and Loya 1979 a). However, Harrison (1985) has concluded after summarizing all of the known coral reproductive characteristics that there are general systematic trends in coral sexuality (hermaphroditism vs. gonochorism) but not in embryonic development (broadcasting vs. brooding). Among Indo-Pacific species, mode of sexuality tends to be conserved within families. It is interesting that several important exceptions to this systematic trend occur among the Caribbean species. Especially notable is the divergence between the Pacific *Porites* spp. which are gonochoric (Kojis and Quinn 1981b; Harrison 1985) and the Caribbean *Porites* spp. which are hermaphroditic (this paper; N. Goreau, personal communication), and also the gonochorism of *Montastrea cavernosa* (this paper) in contrast to the hermaphroditic condition of most other faviids (Kojis and Quinn 1982; Harrison 1985). These reproductive differences may imply important differences in evolutionary selective pressures between the two zoogeographic regions.

Reproductive strategies

Four general reproductive patterns can be proposed for these Caribbean reef corals, (Table 3), based on the patterns exhibited in Table 1, and more may become obvious as other species are studied. These are:

- hermaphroditism followed by broadcast spawning
- hermaphroditism followed by brooding

- gonochorism followed by broadcast spawning
- gonochorism followed by brooding.

These are the same general patterns that have been found for Indo-Pacific species (Harriott 1983 a). The species that are known to follow patterns (a) and (c), under favorable circumstances generally form larger ($\gg 30$ cm diameter), and by implication, longer-lived colonies, while those following patterns (b) and (d) are generally small in size, which implies either a decrease or cessation in growth with age, high adult mortality, or both. The brooding trait is correlated with small colony size and many reproductive cycles per year for other Caribbean species not studied here, such as two species of *Agaricia* (von Moorsel 1983), *Manicina areolata* (Wilson 1888), and the three other species of Caribbean *Porites* (Goreau et al. 1981; N. Goreau, personal communication) as well as for the well studied Indo-Pacific species *Pocillopora damicornis* and *Stylophora pistillata* (Harrigan 1972 and Rinkevich and Loya 1979a, b, respectively).

Can these reproductive patterns be viewed as adaptive to particular situations, and can we venture to guess at the factors that may have led to the evolution of these reproductive strategies?

Disturbance as an evolutionary force

The persistence of a population, and of a species, depends on its birth rate being equal to or greater than its mortality rate. One of the major causes of coral mortality is environmental perturbation, which can occur with varying

Table 3. Summary of reproductive life history strategies, including reproductive (sexuality: hermaphroditic vs. gonochoric; development: broadcasting vs. brooding), growth (large vs. small colony size) and recruitment characteristics (Bak and Engel 1979; Sammarco 1980; Rylaarsdam 1983) of Caribbean reef corals. AC = *Acropora cervicornis*; AP = *A. palmata*; DG = *Dendrogyra cylindrus*; DS = *Diploria strigosa*; FF = *Favia fragum*; MA = *Montastrea annularis*; MC = *Montastrea cavernosa*; MY = *Mycetophyllia ferox*; PA = *Porites astreoides*; SR = *Siderastrea radians*; SS = *S. siderea*

Sexuality	Embryonic Development	Cycles Per Year	Species	Sex Ratio ♀ : ♂	Colony Size	Reported Recruitment Rates
		I	MA, DS AP, AC	N/A		LOW
		$\gg I$	NOT YET REPORTED			
		I	MY	N/A		LOW
		$\gg I$	FF, PA	N/A		HIGH
		I	SS, MC DG	I:I		LOW
		$\gg I$	NOT YET REPORTED			
		I	NOT YET REPORTED			
		$\gg I$	SR	20:I		HIGH

degrees of intensity and frequency and on various spatial scales (Connell 1978; Sousa 1984). Frequent, but localized perturbation can occur in shallow reef zones during "normal" storms, or due to substrate slumping in steep fore-reef areas (Goreau 1959; Connell 1973; Bak and Luckhurst 1980). This contrasts with infrequent but larger scale and more catastrophic perturbations that results from extreme low tides (Glynn 1968; Loya 1976a), hurricanes (Stoddard 1974; Highsmith et al. 1980; Woodley et al. 1981), large scale sea warming events (Glynn 1983), or changes in sea level due to glacial episodes or tectonic activity (Stoddard 1969; Potts 1983; Glynn and Wellington 1983). Those species that specialize to persist in frequently disturbed habitats on coral reefs must be able to either resist the destructive forces or to recolonize those habitats frequently. All species, however, must be able to colonize new habitats after global disturbances if they are to persist over geological time scales. I suggest that these two extremes of disturbance (localized vs. catastrophic) are two factors that have acted in the selection of reproductive strategies of corals.

There are three ways in which coral species could survive frequent physical disturbance. First, they could tolerate high rates of adult mortality by having high rates of local recruitment to replace recently killed adults. High rates of recruitment could be achieved by increasing reproductive effort, or by adopting reproductive characteristics that result in an increase in the proportion of offspring that are successful in recruiting, or both. Secondly, they could resist physical disturbance by constructing skeletons that are resistant to all but the more severe degrees of disturbance. This could be by forming large massive skeletons (e.g. *Montastrea* spp., *Diploria strigosa*, *Siderastrea siderea*), or by forming strong, dense, hydrodynamic skeletons (e.g. *Acropora palmata*). The third means of surviving is by having a high rate of survival of fragments when the colony does break (e.g. *A. cervicornis*, but see Knowlton et al. 1981).

The relationship between these three survival strategies and the reproductive characteristics of the coral species adopting them are discussed next.

Brooding to increase recruitment success

While an organism can always opt for increasing its reproductive effort with the hope of also increasing its reproductive success, this strategy has obvious limitations and drawbacks. More adaptive would be means of increasing its reproductive efficiency (number successful recruits per amount of reproductive effort). I suggest that brooding in corals is a means of increasing both reproductive success and reproductive efficiency. This suggestion is largely speculative and correlative at this point, because there are no good measurements of reproductive effort available with which to make comparisons. However, taken as a whole, brooding, whether by hermaphroditic or gonochoric species, can be seen to be adaptive for situations that require higher local recruitment rates. In

all but one of the species (Table 3) the brooding trait is coupled with many reproductive cycles per year, which means that there are chances for recruitment over most of the year. Brooding bypasses the high larval mortality usually associated with planktonic development and increases the chances of finding appropriate substrate at time of settlement. That the planulae can often settle within hours after release (Lewis 1974; Goreau et al. 1981; personal observation) means that some of the recruitment can take place in the same habitat where the adults have been successful. The smaller number of offspring per brood due to limitations in brooding space is at least partly off-set by having many broods per year. In addition, success in long-distance dispersal by these planulae is not excluded. Richmond (1981) has shown that planulae of *Pocillopora damicornis*, in addition to their lipid stores can feed on zooplankton and receive photosynthetic products from zooxanthellae acquired from their parents during the brooding stage. Support for the adaptiveness of brooding to a life-history depending on high recruitment is the fact that most of the Caribbean species listed as brooders in Table 1 are reported to have high rates of recruitment, while none of those listed as broadcasters had high recruitment rates (Bak and Engel 1979; Rylaarsdam 1983; Sammarco 1980). The determination as to whether brooding results in greater reproductive success by increasing reproductive efficiency or simply because the animals are spending more effort on reproduction must await an accounting of all of the costs of brooding. Preliminary observations are discussed below.

Age versus size

For corals as well as for most other species, growth to a larger size conveys a greater chance of long-term survival (Hughes and Jackson 1985). Therefore, it has been suggested that in species adopting the last two strategies (massive skeleton; high fragment survival), growth should assume a high energetic investment priority during a colony's early life history or just after fragmentation, and reproduction should be delayed until a safe size has been reached (cf., Jackson 1979, Szmant-Froelich 1985).

Connell (1973) reported that several coral species have a delay in the age of first reproduction, but suggested that colony size, and not age, might be the determinant of reproductive ability. Recent reports indicate that small colonies (4–8 cm diameter) of several Australian species either had no gonads or spermaries only (Kojis and Quinn 1982; Harriott 1983), but the ages of their specimens were not specified. Recently completed experiments (Kojis and Quinn 1984; Szmant-Froelich 1985) indicate that colony size is the determining factor for some species. For corals like *Montastrea annularis*, many genetically-old small colonies can be found that are remnants of older, larger colonies, and are the results of partial death of these older colonies (Hughes and Jackson 1980). To determine whether colony size and partial mor-

tality had an effect on the ability of this species to reproduce, fragmentation experiments based on suggestions by Connell (1973) were conducted, and a larger number of smaller colonies were examined for reproductive status (Szmant-Froelich 1985). Most of the colonies smaller than 100 cm² surface area, whether natural or experimental, had either decreased or no gonad development. However, larger experimental colonies and all natural colonies over 300 cm² surface area did have mature gonads. Colonies of a second species, *Mycetophyllia ferox*, smaller than 100 cm² and branches of *Acropora cervicornis* less than 20 cm length did not have gonads either (Szmant, unpublished results). These results indicate that reproductive delay does occur for some species and that it is a function of colony size and not necessarily of tissue age. Similar conclusions have been reached for gorgonians (Wahle 1983).

The effect of size should result in two important effects on coral reproductive schedules. First of all, since most corals exhibit indeterminate growth and reproductive senescence has not been demonstrated in corals, life tables for long-lived corals should show no leveling off or decline of fecundity with increasing size. Therefore, greater colony size in addition to conferring greater chances of colony survival (Hughes and Jackson 1985), also confers a larger reproductive output and a potentially greater fitness (Hughes 1984). The second effect will counterbalance this, in that partial colony mortality that results from disease, physical disturbance, predation, etc., could result in a colony regressing to a smaller, less fecund size class. Partial mortality is a frequent event for many coral species (Hughes and Jackson 1980, 1985). If colony breakage is severe enough, as after a large storm (Highsmith et al. 1980; Knowlton et al. 1981), none of the surviving fragments may be in the reproductive pool. This repression of reproduction may be viewed as adaptive because it allows the reallocation of reproductive resources to growth so as to speed up the reattainment by the fragment of a "safe" size (e.g., similar arguments by MacArthur and Wilson 1967; Pianka 1970; Stearns 1976, 1977). This strategy would be non-adaptive, however, if breakage is too severe or too frequent, since the fragments may never reach reproductive size. Selection under these last conditions should favor initiation of sex at a smaller size.

Brooding versus broadcasting: cost of reproduction

An interesting correlation that drops out of Table 3 is that all of the known Caribbean brooding coral species are most commonly found as small colonies while all of the broadcasters are frequently found as very large colonies.

That brooding could be advantageous to species that are prone to high adult mortality (and thus, by implication, small in size and short-lived) does not explain why all of the larger species studied here (both hermaphroditic and gonochoric) are broadcasters. One possibility is that brooding is only successful as proposed if a large amount

of resource is spent on reproduction. This large expenditure on reproduction could conflict with the demands for a large resource allocation to growth, defense and maintenance, thus making a fast-growing or large-growing and brooding strategy energetically impractical.

However, if reproductive effort, as indicated by annual egg volume production (Table 2), is compared between the various Caribbean corals studied here, it is apparent that the brooders were not the ones with the highest egg volume production. These reproductive effort estimates for the brooders may be low if the adults are providing the embryos with nourishment for respiration and growth during the brooding period, which has not been demonstrated. On the other hand, brooded planulae can use energy resources for growth that would otherwise have been spent on swimming. With these limitations in mind, Table 2 shows that the species with the greatest reproductive effort was a large-sized gonochoric broadcaster and that the two species with the lowest egg volume production estimates were brooders. *Favia fragum*, a small brooding species with twelve reproductive cycles per year, had a similar annual egg volume production to that of *Montastrea annularis*, a large broadcaster with a single reproductive cycle. While it is clear that the question of the cost of brooding must be answered before a complete comparison can be made, it is also clear that assumptions regarding relative reproductive effort of different sized species, or ones with various growth rates, cannot be based solely on the type of embryonic development or on the achieved recruitment rates (e.g., Bak and Engel 1979).

Hermaphroditism versus gonochorism

Selection for outcrossing. Most scleractinian corals examined to date are hermaphroditic (Fadlallah 1983; Harrison 1985; present study); thus, this pattern of distribution of the sexes could possibly be considered the original condition. Furthermore, the pattern of sexuality seems to be consistent, with a few exceptions, within taxonomic families (Harrison 1985). If so, what selective pressures could have resulted in the evolution of gonochoric coral species? When addressing this question, it is important to consider related aspects such as the amount of effort put into the production of each type of gamete, the sex ratio of the population and the way in which fertilization takes place. The adaptiveness of these traits has been well studied for higher plants, and there is an extensive theoretical and empirical literature on the subject. Two selective forces: (1) the selection for increased outcrossing and rates of recombination (Baker 1959, 1984; Maynard Smith 1978; Thompson and Barrett 1981) and (2) sexual selection as a means of optimizing the investment into male and female gametes (Givnish 1982; Willson 1982; Charnov 1983) have been invoked as the reasons for the evolution of self-incompatibility systems and morphological separation of the sexes (monoecy, dioecy, etc.) from self-compatible hermaphroditism. The ability to colonize remote

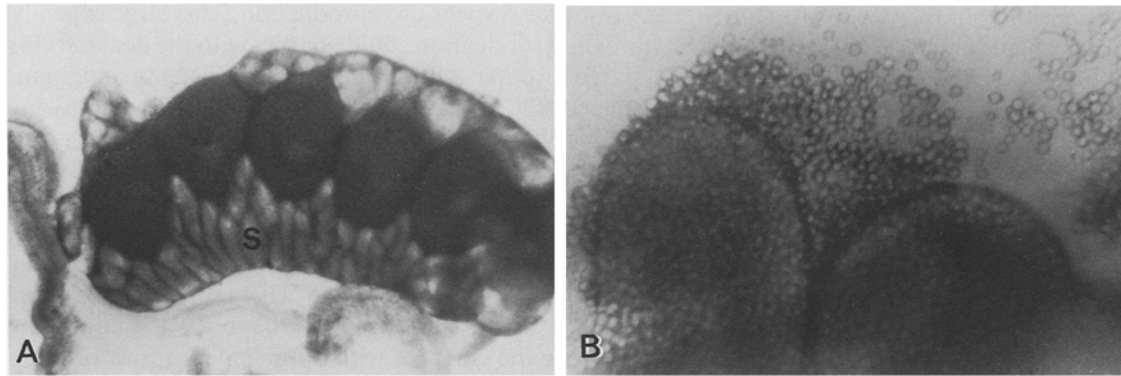


Fig. 2. A Fresh ripe gonad of *Montastrea annularis*. Gonads are weakly attached to the mesentery and thus easily removed from the polyp at this stage. B Ripe oocyte of *M. annularis* squashed open to reveal liquid droplets

areas and maintain small populations has been invoked to explain the retention of self-compatible hermaphroditism. Similar arguments can be made for corals, and these will be discussed next.

For successful reproduction to occur there must be a reasonable amount of fertilization. This depends on gametes of the opposite sex finding each other, and on the sperm being capable of initiating the developmental process in the egg. Hermaphroditic species have the advantage of bearing ripe gametes of both sexes simultaneously and thus increasing the chances of fertilization unless there is self-incompatibility. If self-fertilization results in poor or no development, then self-fertilization will result in wastage of gametes. Complete and partial self-incompatibility have been demonstrated for three Pacific broadcasting coral species, and complete self-compatibility for a fourth (Heyward and Babcock, in preparation). Therefore, while self-fertilization may occur in some species, other broadcasting species would require synchronous spawning for any fertilization to occur.

During spawning, the gonads are released in clusters with the spermaries intact (Harrison et al. 1984; Babcock 1984, personal observation). The eggs are full of lipid droplets (Fig. 2) so that the clusters are extremely buoyant. The spermaries do not break open until the clusters reach the sea surface (Harrison et al. 1984; Babcock 1984), where clusters from different spawning individuals mix. This concentration of gametes on the surface, however, involves a significant risk since they will be extremely susceptible to predation from nocturnally emergent plankton and fish, as well as to factors such as rain squalls (Harrison et al. 1984). Harrison et al. also point out the problem of inter-specific fertilization since they observed over 40 species to spawn at once.

The evolution of self-incompatibility and synchrony in spawning in hermaphroditic species implies that the benefits of outcrossing outweigh the risks described above. Gonochoric species obviously must outcross, and would also be expected to have synchrony in spawning. The presence of gonochoric species supports the concept that outcrossing is an important selective force in some groups of corals.

Selfing versus outcrossing in brooders. If we accept that selection for traits promoting outcrossing has occurred in hermaphroditic broadcasting species and has resulted in the evolution of gonochorism in corals, should we expect equal selection for outcrossing in hermaphroditic brooders? Self-fertilization in plants is often associated with a weedy, opportunistic life history (Baker 1959, and many others), since it allows a single founder seed to colonize an area and create a new population. It is also associated with large trees in tropical rain forests, where the density of each of many species is low, thus making obligate cross-fertilization difficult (Baker 1959). As discussed above, the brooding trait, as observed so far in Caribbean corals, is closely associated with an opportunistic-type life history. Therefore, it would be advantageous for these brooders to be able to achieve a reasonable success of reproduction with self-fertilization. I suggest, however, that while self-fertilization is probably common in these species for the same reasons why selfing is advantageous in higher plants, that there is also a considerable effort made to increase out-crossing.

Brooders such as *Favia fragum* and *Mycetophyllia ferox* had large numbers of spermaries, that often outnumbered the eggs in number and in volume by a factor of five to ten, based on histological observations (Fig. 3). This would appear to be extremely wasteful if self-fertilization was common, and rather implies an intense effort to out-cross similar to that of trees that produce enormous quantities of pollen (Baker 1959; Williams 1975). The basal position of the ripe eggs and distal location of the ripe spermaries within the polyp mesenteries would also reduce the chance of self-fertilization (Szmant-Froelich et al. 1985). These species may release sperm attractants (Miller 1982), which could increase the chances of cross-fertilization if the attractant were released at different times from the sperm.

Sex ratios of gonochoric species. Success in fertilization of a gonochoric species will depend on the population density of the species and its sex ratio. *Montastrea cavernosa* and *Siderastrea siderea* are both common species on Puerto Rican reefs, and if spawning is synchronous, their

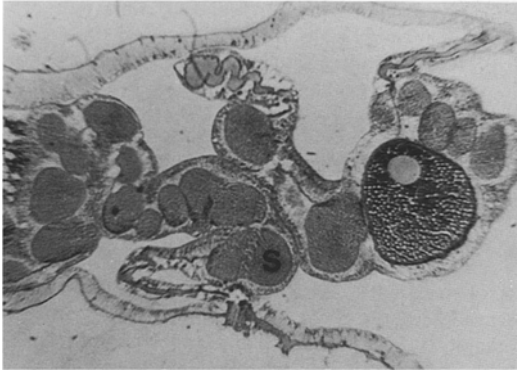


Fig. 3. Longitudinal section through a polyp of *Mycetophyllia ferox* showing large number of spermaries per oocyte

gametes should have a reasonable chance of finding each other. *Dendrogyra cylindrus*, on the other hand, is rare on many reefs. At the La India sampling site, a single large male colony occurred within an area of over 4,000 m², while on El Mario reef there are about 30 large colonies within a similar area. Small colonies are unusual. Reproductive success of this species could therefore be severely limited by too small a population size, and could mean that this species is in danger of local extinction. While reported to occur in Florida, Belize and the West Indies, it is conspicuous by its absence from present day Panamanian coral reefs although recent fossils (1000 years old) have been found in the area (Glynn, personal communication).

Preliminary estimates of the sex ratios of the three broadcasting gonochoric species based on small arbitrary samples (40 colonies) of two populations show that the ratios are very close to one, as would be predicted for resource allocation with random mating (Charnov 1983). Interestingly, a sample of 25 colonies from a small population of *Siderastrea radians*, a gonochoric species which differs from its congener *S. siderea* in that the female colonies brood, resulted in a ratio of 22 females to 3 males. The habitat and type of morphology of this species makes it unlikely that the larger number of females could be explained by asexual reproduction by a female colony (Harrriott 1983). Since only the female colonies of *S. radians* brood and brooding space within each female polyp is limited, in a population with an equal number of male and female colonies, many males may not succeed in breeding very often. The high female to male sex ratio would be expected to evolve in the population (e.g., a mutation which favored an increase in the proportion of females among the offspring), because genomes producing a large number of female offspring will be more successful in transmitting their genes (Maynard Smith 1978; Charnov 1983). Female colonies might also be expected to release sperm attractants to increase the success of internal fertilization (Miller 1982). These observations need to be verified by sampling other populations; they were especially unexpected since *S. radians* was earlier reported to be hermaphroditic (Duerden 1902).

Conclusions

Based on their reproductive traits, the eleven species of Caribbean corals studied here can be grouped into four reproductive strategies. Broadcast spawning, whether by hermaphroditic or gonochoric species is exhibited by species which often attain large colony size, and may be an adaptation for increased outcrossing between individuals that can be very long-lived, due either to resistance to most causes of mortality or to vegetative reproduction. Individuals of these species appear to delay reproduction until a "safe" colony size has been attained, as this increases the chances of survival of the individual, and they lose their ability to reproduce sexually if their colony size decreases below a certain level due to partial colony mortality or fragmentation. They also tend to have a single reproductive event per year.

Brooding of embryos to an advanced larval stage appears to somehow increase the chances of recruitment by that species, possibly by providing the larvae suitable habitat for settlement immediately upon release, or by producing larger, more fit planulae, or even by increasing the probability that successful fertilization takes place. Many of these brooders are smaller-sized species considered to be opportunistic or weedy (Loya 1976b; Bak and Engel 1979; Bak and Luckhurst 1980), and are characteristic of reef zones frequently impacted by storms, low tidal exposures or other forms of physical disturbance. Therefore, brooding, with the resulting higher recruitment rates, appears to be an adaptation evolved by coral species that suffer frequent and early colony mortality. Colonies of these species begin sexual reproduction within a few years of settlement, but it is not known whether partial mortality or fragmentation alters their reproductive capacity.

It is interesting to note that while the Caribbean *Porites* species form relatively small colonies (with the exception of *P. porites* which can vegetatively reproduce to cover extensive areas; Glynn 1973) and are hermaphroditic brooders, most Indo-Pacific species of *Porites* are gonochoric, massive in size and annual broadcasters (Kojis and Quinn 1981b; B. Kojis, personal communication). The one Pacific species known to brood, *P. murrayensis*, forms small encrusting colonies and is relatively short-lived (Kojis and Quinn 1981b).

Brooding by gonochoric species, if the brooding is done only by females, should result in a sex ratio skewed towards a higher proportion of females, such as was found for *Siderastrea radians*. The sex ratios of gonochoric broadcasters, however, should be close to one, as was found for the three species studied here (Table 1).

These four reproductive strategies are certainly not the only ones possible, and I am sure that more will be described as more species are studied. One type of pattern not seen in the Caribbean so far is that exhibited by *Goniastrea favulus*, which spawns heavy, sticky eggs that undergo external but demersal development (Kojis and Quinn 1981a, b). In addition, while not exhibited by any

of the species studied here, there should exist large-sized coral species that brood their larvae and that possibly have higher recruitment rates. The purpose of this paper is not to limit the possibilities as to the types of reproductive patterns expected or possible in scleractinian corals, but to synthesize the information in hand and to encourage others to consider the adaptive and evolutionary aspects of observed coral reproductive patterns.

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